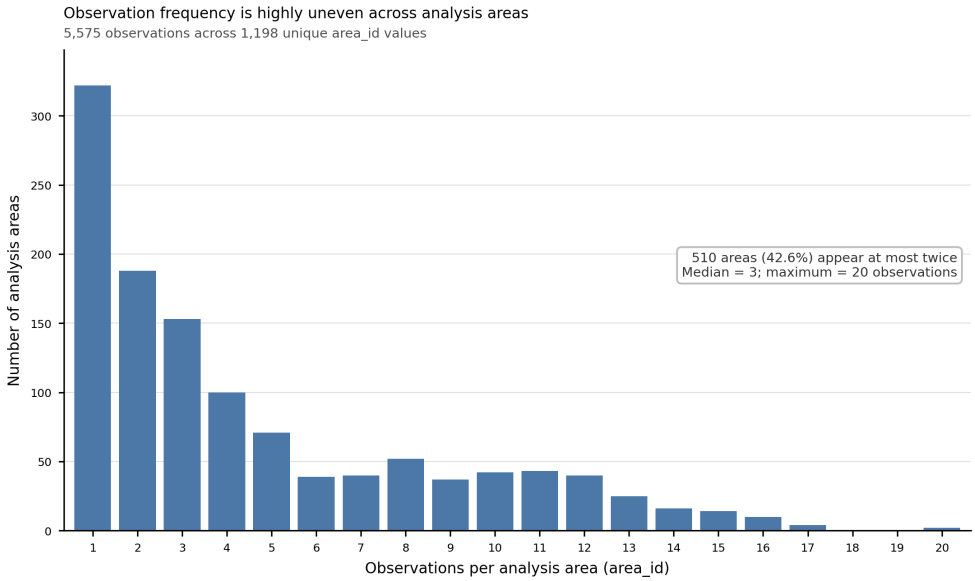
Predicting Acute Food Insecurity with Machine Learning

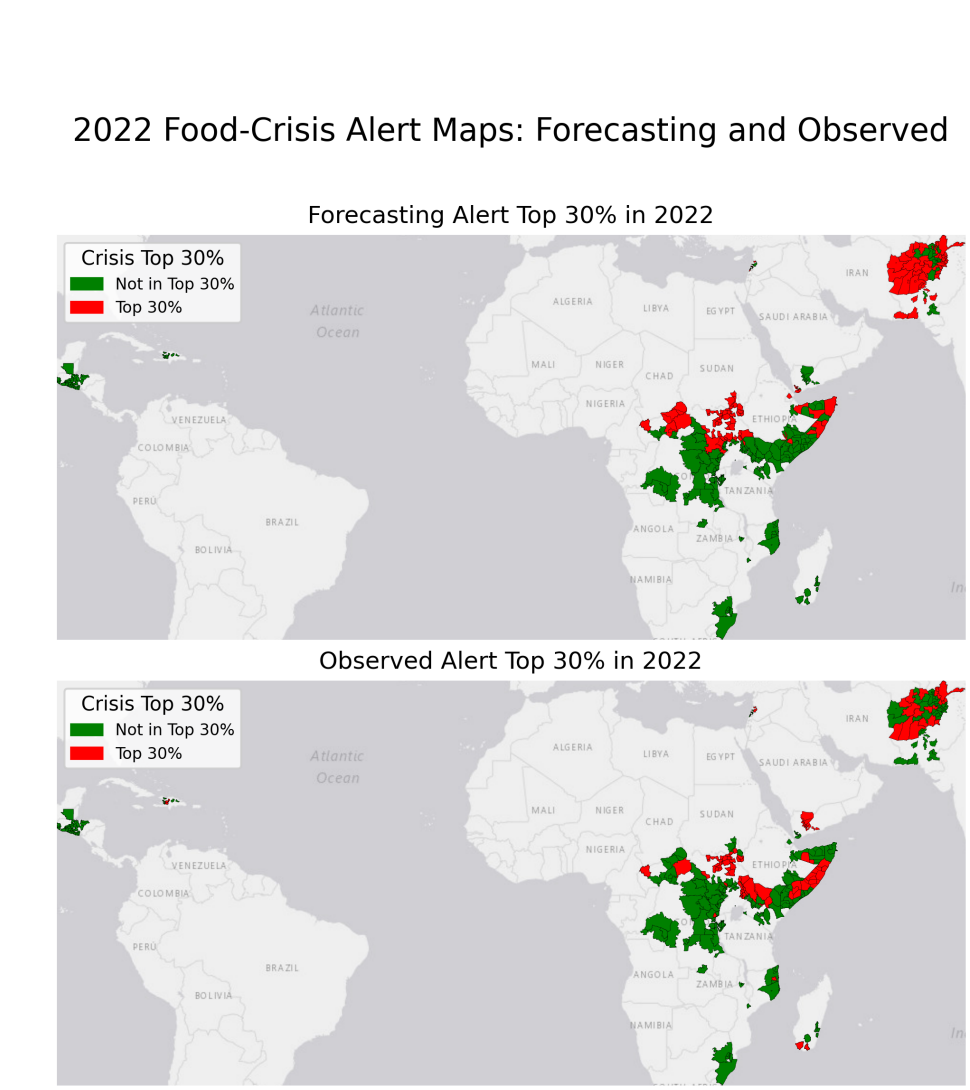
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Supplementary Figures

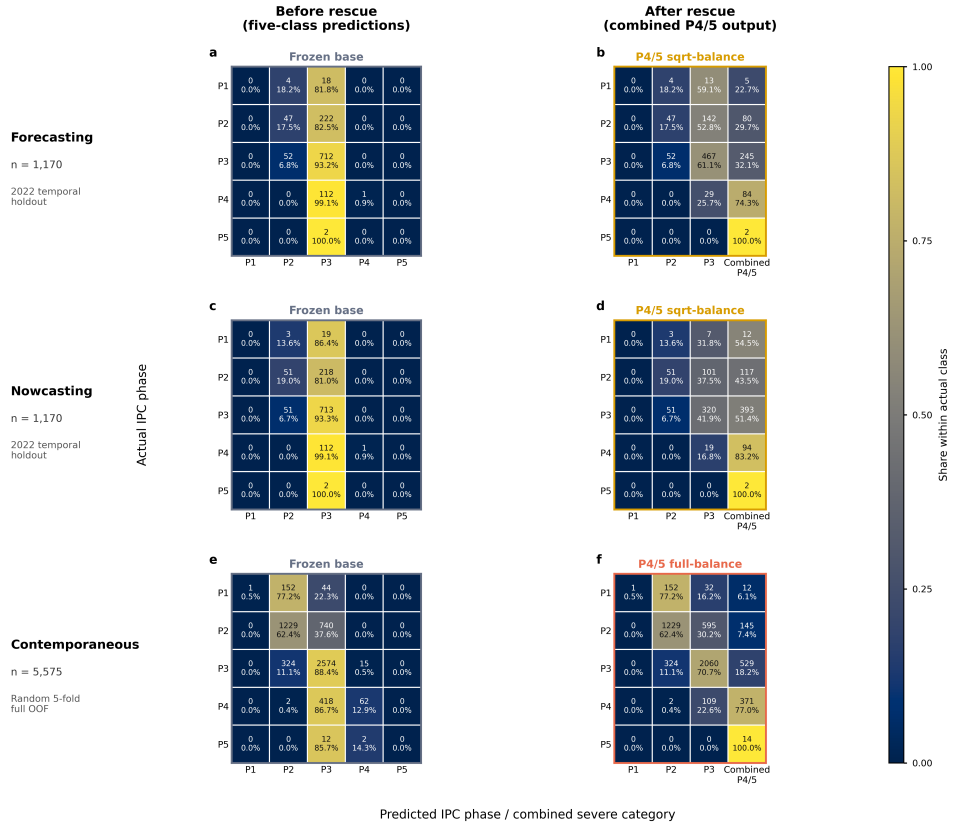


Supplementary Fig. 1 Uneven Observation Frequency Across Analysis Areas
Notes: The histogram shows the number of observations available for each analysis area. The horizontal axis indicates the number of observations associated with an individual area, and the vertical axis indicates the number of unique analysis areas with that observation frequency. The sample contains 5,575 observations from 1,198 unique analysis areas. Observation frequency is strongly unbalanced: 510 areas (42.6%) appear no more than twice, the median area appears three times, and the maximum number of observations for a single area is 20.

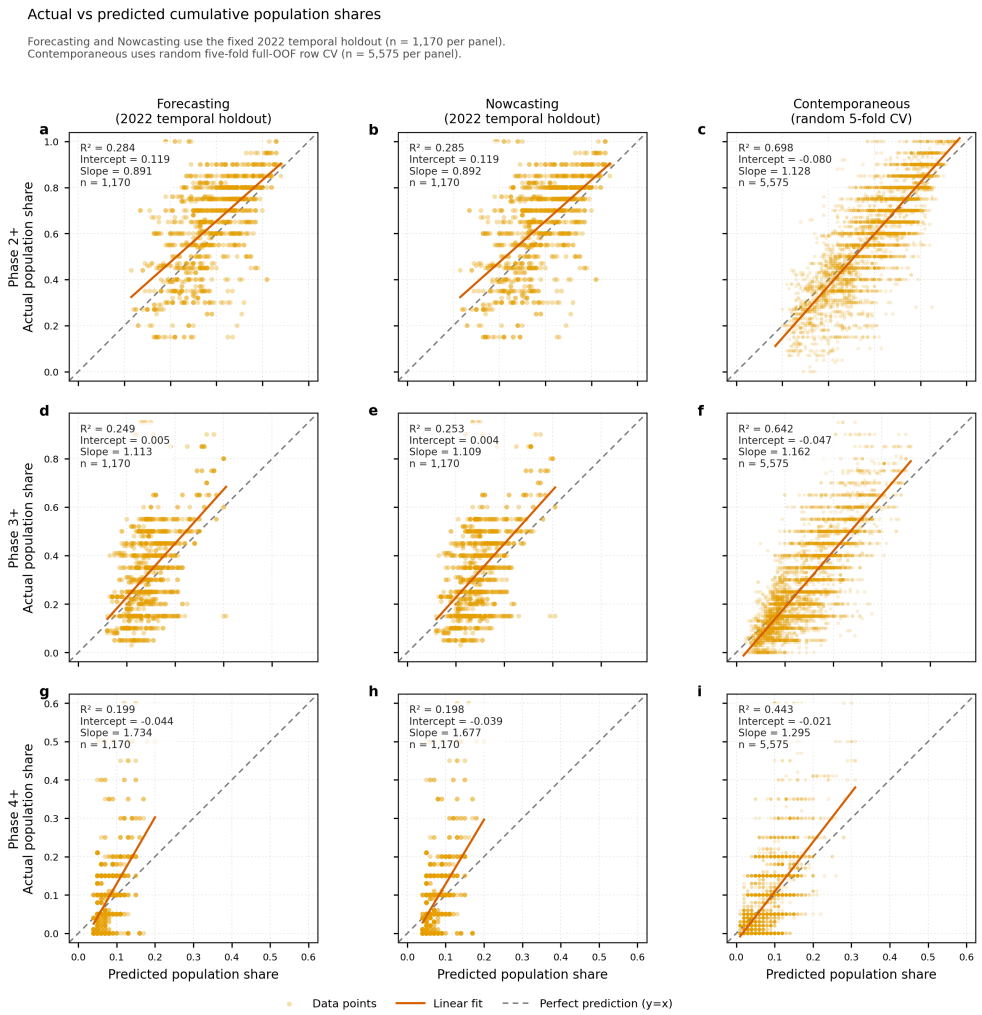


Supplementary Fig. 2 Top 30% Predicted and Observed Food-Insecure Areas by Population Share in Food Crisis

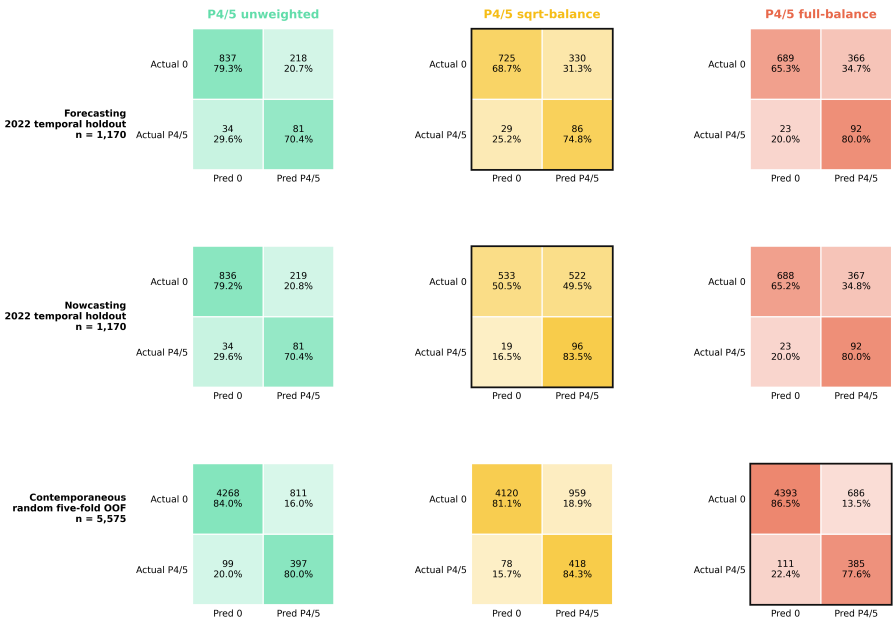
Notes: The maps compare the spatial distribution of analysis areas in the highest 30% of the population share experiencing food crisis. Red areas in the upper panel indicate locations predicted, 12 months in advance, to fall within the top 30%, while red areas in the lower panel indicate locations in the top 30% based on observed outcomes. The remaining evaluated areas are shown in green. Using the same threshold facilitates comparison of the geographic concentration and correspondence between predicted and observed high-risk areas.



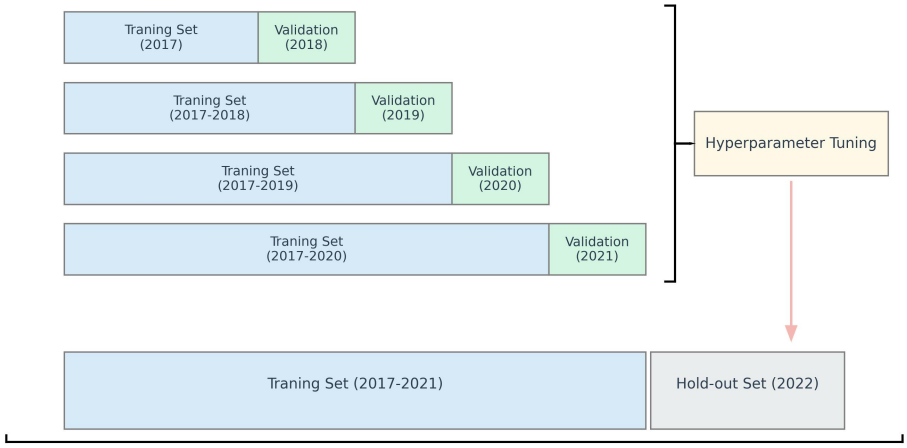
Supplementary Fig. 3 IPC Predictions Before and After Combined Phase 4/5 Rescue
Notes: For the contemporaneous panels, actual phases are reconstructed from the observed cumulative population shares using the model's 20% threshold rule, rather than taken from the source-table phase labels used in Fig. **??**. Rows show forecasting (a–b), nowcasting (c–d), and contemporaneous modeling (e–f). Left panels retain five predicted IPC classes. Right panels display four predicted categories (P1, P2, P3, combined P4/5), summing the stored P4 and P5 columns. Both sides retain all five actual classes and identical observations. Rescue promotes eligible baseline P3 predictions to combined P4/5; it does not distinguish P4 from P5 or establish improved full five-class classification. Cells report counts and percentages within each actual phase; shading represents the same within-class share. Forecasting and nowcasting use the square-root-balance policy on the 2022 temporal holdout ($n = 1{,}170$ each). Contemporaneous uses the full-balance policy and seed-0 random five-fold row-level OOF predictions ($n = 5{,}575$), with its threshold selected from pooled OOF results.



Supplementary Fig. 4 Actual and Predicted Cumulative IPC Population Shares
Notes: These diagnostics assess cumulative population-share predictions, not probabilities that an area experiences a food crisis. Columns show forecasting, nowcasting, and contemporaneous estimates. Rows show IPC Phase 2+ (panels a–c), Phase 3+ (panels d–f), and Phase 4+ (panels g–i). Each point represents one observation, with predicted share on the horizontal axis and observed share on the vertical axis. Orange lines regress observed shares on predicted shares; dashed gray lines show equality ($y = x$). Intercept 0 and slope 1 are the linear-calibration reference. Reported R^2 compares the original share predictions with observations, rather than measuring the orange line's fit. Forecasting and nowcasting use the 2022 temporal holdout ($n = 1{,}170$ per panel); contemporaneous estimates use random five-fold OOF predictions ($n = 5{,}575$ per panel).



Supplementary Fig. 5 Binary Confusion Matrices for the Phase 4/5 Rescue Models
Notes: The matrices summarize binary classification performance for identifying observations in IPC Phase 4 or 5 under three class-weighting policies: unweighted, square-root balance, and full balance. Rows correspond to forecasting, nowcasting, and contemporaneous models, while columns correspond to the three weighting policies. "Actual 0" denotes observations below Phase 4, and "Actual P4/5" denotes observations in Phase 4 or 5. Each cell reports the observation count and the percentage within the corresponding actual-class row. Black borders identify the primary policy selected separately for each modeling task using out-of-fold (OOF) results. Forecasting and nowcasting are evaluated on the fixed 2022 temporal holdout ($n = 1{,}170$ each), whereas the contemporaneous model uses seeded random five-fold, row-level OOF predictions ($n = 5{,}575$), with its threshold selected from the pooled OOF predictions.



Supplementary Fig. 6 Temporal Validation for Forecasting Hyperparameter Selection
Notes: Forecasting hyperparameters were selected using four expanding-window temporal validation folds restricted to 2017–2021, adapted from [**?**]. Training began with 2017, followed by validation in 2018; subsequent folds expanded training annually and validated in 2019, 2020 and 2021. The 2022 observations were excluded from every tuning fold. For final forecasting and nowcasting evaluation, observations from 2017–2021 formed the training set and 2022 observations formed the fixed held-out test set. Nowcasting and contemporaneous models were evaluated with fixed hyperparameters, without separate hyperparameter searches.

Supplementary Tables

Supplementary Table 1 Model Performance Relative to Benchmark Models

Model	Accuracy	Sensitivity	Precision	R^2
Panel A: Persistence baseline				
Forecasting/Nowcasting	0.510	0.612	0.853	–
Panel B: Ordered Probit				
Forecasting	0.491	0.613	0.793	–
Nowcasting	0.490	0.564	0.825	–
Contemporaneous	0.654	0.792	0.840	–
Panel C: OLS				
Forecasting	0.533	0.689	0.784	-0.796
Nowcasting	0.545	0.694	0.794	-0.930
Contemporaneous	0.639	0.742	0.888	0.443
Panel D: ML comparable to Panel A: persistence baseline				
Forecasting	0.672	0.918	0.804	0.243
Nowcasting	0.675	0.914	0.811	0.254
Panel E: ML comparable to Panels B and C				
Forecasting	0.650	0.941	0.775	0.249
Nowcasting	0.654	0.942	0.777	0.253
Contemporaneous	0.693	0.904	0.797	0.642

Notes: This table compares the machine-learning models with persistence, ordered probit, and OLS benchmarks. Accuracy evaluates overall IPC phase classification; sensitivity and precision evaluate food-crisis classification; while out-of-sample R^2 evaluates prediction of the population share in food crisis (Phase 3 or above). The corresponding ML results in Panel D are obtained by refitting the ML models on the comparably restricted sample.

Supplementary Table 2 Model Performance for Alternative Spatial Strategies

Model	Accuracy	Sensitivity	Precision	R^2
Panel A: Exclusion of latitude and longitude				
Forecasting	0.674	0.954	0.789	0.217
Nowcasting	0.676	0.954	0.794	0.223
Contemporaneous	0.686	0.906	0.789	0.628
Panel B: 5-nearest-neighbor spatial averages				
Forecasting	0.671	0.945	0.796	0.134
Nowcasting	0.674	0.944	0.803	0.141
Contemporaneous	0.700	0.912	0.804	0.663
Panel C: 200-km spatial averages				
Forecasting	0.666	0.935	0.802	0.131
Nowcasting	0.673	0.936	0.804	0.136
Contemporaneous	0.698	0.906	0.804	0.680

Notes: This table reports robustness checks using alternative spatial specifications: excluding latitude and longitude, replacing coordinates with five-nearest-neighbor spatial averages of predictors, and using spatial averages within 200 km. Accuracy evaluates overall IPC phase classification; sensitivity and precision evaluate food-crisis classification; while out-of-sample R^2 evaluates prediction of the population share in food crisis (Phase 3 or above). For the five-fold area-level cross-validation, reported performance metrics are the arithmetic means of the corresponding fold-specific metrics, rather than metrics recomputed from pooled out-of-fold predictions.

Supplementary Table 3 Generalization to New Geographic Areas

Model	Accuracy	Sensitivity	Precision	R^2
Panel A: Leave-One-Country-Out				
Forecasting	0.658	0.960	0.773	-0.004
Nowcasting	0.665	0.964	0.781	0.008
Contemporaneous	0.583	0.941	0.671	0.136
Panel B: Five-Fold Area-Level Cross-Validation				
Forecasting	0.649	0.908	0.798	0.165
Nowcasting	0.656	0.914	0.802	0.170
Contemporaneous	0.830	0.923	0.745	0.718

Notes: This table reports model performance under two validation exercises designed to assess generalization to previously unseen geographic areas. Panel A reports leave-one-country-out results and Panel B reports five-fold area-level cross-validation results. Accuracy evaluates overall IPC phase classification; sensitivity and precision evaluate food-crisis classification; while out-of-sample R^2 evaluates prediction of the population share in food crisis (Phase 3 or above).

Supplementary Table 4: Base Variables and Data Sources by Variable Group

Variable Name	Dataset Source
Panel A: Physical Geography	
Elevation	ERA5-Land
Ruggedness	Nunn, Nathan, and Diego Puga (2012)
Slope	ESA
Distance to rivers	World Bank and Andreadis et al. (2013)
Panel B: Conflict	
Event count of battles	ACLED
Event count of explosions	ACLED
Event count of violence against civilians	ACLED
Fatalities of battles	ACLED
Fatalities of explosions	ACLED
Fatalities of violence against civilians	ACLED
Reverse distance weight of fatalities in neighboring 5 stations	ACLED
Reverse distance weight of fatalities in neighboring 10 stations	ACLED
Reverse distance weight of event counts in neighboring 5 stations	ACLED
Reverse distance weight of event counts in neighboring 10 stations	ACLED
Panel C: Weather Conditions	
Rainfall	CHIRPS
2m temperature	ERA5-Land
Standard deviation of 2m temperature	ERA5-Land
Surface solar radiation downwards	ERA5-Land
Standard deviation of surface solar radiation downwards	ERA5-Land
Panel D: Food Prices	
Distance to nearest food market	FAO FPMA
Food price index (reference-price normalized)	FAO FPMA
WFP price change percentage (monthly)	WFP
WFP spatial food price dispersion(monthly)	WFP

Continued on next page

Supplementary Table 4 (continued)

Variable Name	Dataset Source
Panel E: Economic	
Estimated population	IPC
CPI (2010 = 100)	WBG (FP.CPI.TOTL)
GDP per capita	WBG
Gini coefficient	WBG
Control of corruption: percentile rank	WBG
LSCI infrastructure score	WBG (IS.SHP.GCNW.XQ)
VIIRS nightlight mean	VIIRS
Standard deviation of VIIRS nightlight	VIIRS
Market access	Market Access (Weiss, 2015)
Panel F: Agriculture and Land Productivity	
Agricultural ecological zone	Tricht et al. (2023)
GOSIF GPP	GOSIF-GPP
Standard deviation of GOSIF GPP	GOSIF-GPP
Soil volumetric water	ERA5-Land
Standard deviation of soil volumetric water	ERA5-Land
Enhanced vegetation index	NASA
Cropland mask	ASAP (European Commission JRC)
Rangeland mask	ASAP (European Commission JRC)
Volumetric fraction of coarse fragments	ISRIC
Total nitrogen	ISRIC
Soil pH value	ISRIC
Soil organic carbon content in the fine earth fraction	ISRIC
Cation exchange capacity of the soil	ISRIC

Supplementary Notes

We report the hyperparameter search space, temporal validation procedure, model-selection criterion, and final effective parameter values here. We also describe the treatment of missing predictor values in the ML model and in the two benchmark models, ordered probit and OLS.

Hyperparameter tuning

Hyperparameter tuning was performed for forecasting. Nowcasting and contemporaneous models were subsequently evaluated with fixed hyperparameters, without separate hyperparameter searches. The forecasting search exhaustively enumerated the Cartesian product of the eight hyperparameter grids shown below; it was not based on random sampling. Tuning was conducted in two stages. We first selected the parameters for the Phase 3+ cumulative target. In the second stage, the Phase 3+ component was held fixed while parameter settings were selected for the Phase 2+, Phase 4+, and Phase 5 cumulative targets. We prioritized Phase 3+ because it defines food-crisis conditions requiring urgent response under the IPC framework, making both the population share in Phase 3+ and the identification of areas classified as Phase 3+ particularly policy relevant. All 11,664 candidate configurations were evaluated in each stage, corresponding to 23,328 candidate-configuration evaluations across the two searches. We searched all the combinations as it was computationally feasible in this situation and more efficient search strategies could be employed for larger search spaces.

Temporal validation and selection criterion

Each candidate configuration was evaluated using four expanding-window temporal validation folds: 2017 was used to train the first fold and 2018 for validation; 2017–2018 were used for training and 2019 for validation; 2017–2019 were used for training and 2020 for validation; and 2017–2020 were used for training and 2021 for validation. The 2022 data were reserved exclusively for final holdout evaluation. Candidate configurations were ranked by their mean validation score across the four folds:

$$\text{Selection score} = \text{Accuracy} + \text{Precision} + 5 \times \text{Sensitivity} + R^2_{\text{Phase 3+}}$$

The score combines accuracy, precision, sensitivity, and the R^2 for the Phase 3+ population share. We assign greater weight to sensitivity because failing to identify true food-crisis cases is of particular concern in an early-warning setting.

Final Effective Parameter Values

Supplementary Table 5 reports the hyperparameter search space. The final effective parameter values used to produce the reported predictions are listed below.

Forecasting model. For the final forecasting model, the Phase 2+ cumulative target used max_depth = 11, n_estimators = 200, learning_rate = 0.10, subsample = 1.0,

colsample_bytree = 0.5, gamma = 0, min_child_weight = 0, and scale_pos_weight = 1. The Phase 3+, Phase 4+, and Phase 5 cumulative targets used the same parameter set: max_depth = 9, n_estimators = 200, learning_rate = 0.10, subsample = 0.5, colsample_bytree = 0.7, gamma = 0.1, mi_child_weight = 0, and scale_pos_weight = 1.

Nowcasting model. The two layers of the nowcasting model used the same parameter settings as the forecasting model.

Contemporaneous model. The contemporaneous model used a common parameter set for all four cumulative targets (Phase 2+, Phase 3+, Phase 4+, and Phase 5): max_depth = 3, n_estimators = 200, learning_rate = 0.10, subsample = 1.0, colsample_bytree = 0.7, gamma = 0.1, min_child_weight = 0, and scale_pos_weight = 1.

All other parameters were left at the corresponding corresponding python xgboost library defaults(xgboost.readthedocs.io). The released reproducible implementation fixes the estimator random state at 0.

Missing values

Our models include 172 predictors, many of which contain missing values. Patterns of missingness may vary systematically across countries, data sources, and conflict contexts.[1] XGBoost handles missing predictor values natively by learning a default direction for missing observations at each tree split. In contrast, OLS and ordered probit require finite design matrices, so their performance may depend on the chosen missing-data treatment.

For the ordered probit and the OLS models reported in Supplementary Table 1, we use feature-wise median imputation estimated from the training data only, followed by standardization using the corresponding training-sample moments. These preprocessing parameters are fixed after training and applied during inference.

We conduct two additional missing-data robustness checks. First, we re-estimate the OLS and ordered probit benchmarks after adding a binary missingness indicator for each predictor; these results are reported in Panels A and B in Supplementary Table 6. Second, we repeat the model comparisons after excluding the 10% most-missing predictors and, separately, the three countries with the highest missingness rates. These results are reported in Panels C–F, with the corresponding ML results shown in Panels G and H of Supplementary Table 6.

[1] Using all 5,575 observations, we find that conflict-affected observations have, on average, 2.92 fewer missing forecasting predictors and 2.11 fewer missing nowcasting predictors than non-conflict observations. Inference clustered at the area level indicates that these differences are statistically significant, although modest in magnitude.

Supplementary Table 5
Hyperparameter Search Space

Hyperparameters	Search Space
max_depth	3, 7, 9, 11
n_estimators	200, 500, 800
learning_rate	0.01, 0.05, 0.08, 0.10
subsample	0.5, 0.7, 1.0
colsample_bytree	0.5, 0.7, 1.0
gamma	0, 0.1, 0.2
min_child_weight	0, 1, 2
scale_pos_weight	1, 2, 3

Supplementary Table 6: Robustness Check: Model Performance Relative to Benchmark Models

Model	Accuracy	Sensitivity	Precision	R^2
Panel A: Ordered Probit (+ missingness indicators)				
Forecasting	0.519	0.642	0.798	–
Nowcasting	0.551	0.753	0.795	–
Contemporaneous	0.663	0.805	0.845	–
Panel B: OLS (+ missingness indicators)				
Forecasting	0.544	0.712	0.792	-0.571
Nowcasting	0.603	0.877	0.760	-0.496
Contemporaneous	0.646	0.887	0.752	0.466
Panel C: Ordered Profit (drop 10% most-missing predictors)				
Forecasting	0.498	0.610	0.801	
Nowcasting	0.507	0.613	0.804	
Contemporaneous	0.647	0.785	0.836	
Panel D: OLS (drop 10% most-missing predictors)				
Forecasting	0.534	0.689	0.788	-0.743
Nowcasting	0.546	0.699	0.801	-0.756
Contemporaneous	0.642	0.888	0.748	0.439
Panel E: Ordered Probit (drop 3 most-missing countries)				
Forecasting	0.489	0.608	0.794	
Nowcasting	0.488	0.544	0.843	
Contemporaneous	0.655	0.843	0.792	
Panel F: OLS (drop 3 most-missing countries)				
Forecasting	0.531	0.686	0.783	-0.792
Nowcasting	0.543	0.689	0.796	-0.931
Contemporaneous	0.642	0.890	0.744	0.443
Panel G: ML comparable to Panel A, B, C & D				
Forecasting	0.650	0.941	0.775	0.249
Nowcasting	0.654	0.942	0.777	0.253
Contemporaneous	0.693	0.904	0.797	0.642
Panel H: ML comparable to Panel E & F				
Forecasting	0.676	0.950	0.794	0.294
Nowcasting	0.679	0.942	0.803	0.303
Contemporaneous	0.690	0.905	0.794	0.637

Notes: This table compares the machine-learning models with persistence, ordered probit, and OLS benchmarks. Accuracy evaluates overall IPC phase classification; sensitivity and precision evaluate food-crisis classification; while out-of-sample R^2 evaluates prediction of the population share in food crisis (Phase 3 or above).